

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10: 1.2 INDICES AND LOGARITHMS

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Write your answers in the space under each prompt. Use full sentences and show every step of your working. Name the index law or logarithm law you use at each step. Keep referring to the Kericho tea farm: a single plant gives 2^3 grams of processed tea per harvest cycle, and the estate has 2^{10} plants. Use a four-figure table where a question asks for one, and give answers to a sensible degree of accuracy. Your final answer should link back to the driving question: How can a Kericho tea farmer, or any Kenyan scientist, engineer or business person, use the mathematics of indices and logarithms to make sense of, and calculate with, numbers that are too large or too small to handle by ordinary arithmetic?

Part 1: Indices as a compact language for the tea farm

A Kericho tea plant gives 2^3 grams of processed tea per harvest cycle. The estate has 2^{10} plants. (a) Write 2^3 as repeated multiplication and give its value in grams. (b) Complete this sentence and explain why it is true: when the exponent of 2 goes up by 1, the value _____. What happens when the exponent goes down by 1? (c) Write the fraction of the estate's plants that one single plant represents, first using a reciprocal and then using a negative index. (d) Explain why a negative index does not mean 'negative tea'.

(a) $2^3 = 2 \times 2 \times 2 = 8$ grams per plant. (b) When the exponent of 2 goes up by 1, the value doubles, because we multiply by one more factor of 2 (for example $2^3 = 8$ and $2^4 = 16$). When the exponent goes down by 1 the value halves ($2^3 = 8$, $2^2 = 4$). This is why index notation is so compact: a small change in the exponent gives a big change in the number. (c) One plant out of 2^{10} plants is the fraction $1/2^{10}$. Using the negative index rule $a^{-n} = 1/a^n$, this is 2^{-10} . This links reciprocals to negative indices: the reciprocal of a is $1/a$, and $a \times (1/a) = 1$. (d) A negative index signals a reciprocal, not a negative quantity. 2^{-10} is a very small positive fraction ($1/1024$), so it describes a small share of the estate, not a negative amount of tea.

Part 2: Deriving and using the laws of indices

(a) The farm manager wants the total production of the estate, which is 2^3 grams per plant for

(a) Total = $2^3 \times 2^{10}$. Expanded, this is $(2 \times 2 \times 2) \times (2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2)$. Counting gives $3 + 10 = 13$ factors of 2, so the total is 2^{13} . The product rule is $a^m \times a^n = a^{m+n}$: when multiplying powers of the same base, the exponents add. Notice that we added exponents to get the answer to a multiplication problem. This is the idea behind the logarithm trick. (b) By the quotient rule,

2¹⁰ plants. Write the total as a product of two powers. Expand both powers into repeated multiplication, count the factors of 2, and write the total as a single power of 2. State the index law you have found. (b) Use the quotient rule with equal exponents (for example $5^2 \div 5^2$) to explain why $a^0 = 1$ for any non-zero a. (c) Use the power rule to simplify $(2^3)^2$ and check your answer by writing out all the factors. (d) Explain what $2^{1/2}$ means and why it is not the same as $2 \div 2$.	$a^m \div a^n = a^{m-n}$. When $m = n$, $5^2 \div 5^2 = 5^{2-2} = 5^0$. But any number divided by itself is 1, so $5^2 \div 5^2 = 25 \div 25 = 1$. Therefore $5^0 = 1$, and the same reasoning works for any non-zero a, so $a^0 = 1$. (c) The power rule is $(a^m)^n = a^{mn}$, so $(2^3)^2 = 2^6 = 64$. Check: $(2^3)^2 = 8 \times 8 = 64$, which is six factors of 2 ($2 \times 2 \times 2 \times 2 \times 2 \times 2$). (d) By the fractional index rule, $a^{1/n}$ is the nth root of a, so $2^{1/2} = \sqrt{2} \approx 1.414$. It is the number which, multiplied by itself, gives 2, since $(2^{1/2})^2 = 2^1 = 2$. It is not $2 \div 2 = 1$.
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Part 3: Logarithms as exponents, and the antilogarithm	
(a) State the definition of the common logarithm in terms of a power of 10. (b) Use the definition to write $\log_{10} 1,000$ and $\log_{10} 0.01$ as numbers, and explain how the second connects to the pH of lemon juice, where $[H^+] = 10^{-2}$ mol/L. (c) Without using the table, state the characteristic of $\log_{10} 3,500$ and justify it using powers of 10. (d) Explain, with reference to the definition, what the antilogarithm of 7.2304 means and why the antilog table is really a table of powers of 10. Name the four steps for finding an antilogarithm from a table, using the M-Pesa figure of about 17,000,000 transactions per day as your example.	(a) $\log_{10} x = n$ if and only if $10^n = x$. The logarithm is the exponent to which 10 must be raised to give x. (b) $10^3 = 1,000$, so $\log_{10} 1,000 = 3$. $10^{-2} = 0.01$, so $\log_{10} 0.01 = -2$. For lemon juice $[H^+] = 10^{-2}$ mol/L, so the exponent -2 is the logarithm of the concentration, and the pH is its magnitude, 2. A change of 1 unit in the exponent is a tenfold change in concentration, just as a change of 1 in the exponent of 2 doubles the tea harvest. (c) $10^3 \leq 3,500 < 10^4$, so $\log_{10} 3,500$ lies between 3 and 4 and the characteristic is 3. (d) The antilogarithm of x is 10^x, so the antilog of 7.2304 is $10^{7.2304}$, which is the number whose log is 7.2304. Since the antilog table gives powers of 10, it is the logarithm table read backwards. The four steps are: (1) separate the characteristic (7) from the mantissa (0.2304); (2) find the mantissa in the antilog table using the row for the first two decimal digits (23) and the column for the third digit (0); (3) add the mean-difference correction for the fourth digit (4); (4) use the characteristic to place the decimal point, which gives a number with 8 digits before the decimal point, about 17,000,000.

Part 4: Why adding logarithms gives a product

The fellow farmer at the Kericho training session found the total tea harvest from 2^3 g per plant and 2^{10} plants by using a four-figure log table. (a) Use the index law you met in Part 2 to explain why $\log(a \times b) = \log a + \log b$. Also state what the quotient law is and where it comes from. (b) Using the four-figure table values $\log 8 = 0.9031$ and $\log 1,024 = 3.0103$, carry out the farmer's three steps to find the total harvest in grams. Show each step, then compare your answer with 2^{13} . (c) Why is the farmer's method faster than long multiplication? What does the table do that your brain finds hard?

(a) Write $a = 10^p$ and $b = 10^q$, so that $\log a = p$ and $\log b = q$. Then $a \times b = 10^p \times 10^q = 10^{(p+q)}$ by the product rule for indices. By the definition of a logarithm, $\log(ab) = p + q = \log a + \log b$. In the same way, $10^p \div 10^q = 10^{(p-q)}$, so $\log(a/b) = \log a - \log b$. Logarithms are exponents of 10, and multiplying powers of 10 adds the exponents. (b) Step 1: find the logs: $\log 8 = 0.9031$ and $\log 1,024 = 3.0103$. Step 2: add them: $0.9031 + 3.0103 = 3.9134$. Step 3: find the antilog of 3.9134. The characteristic 3 means the answer has 4 digits before the decimal point, and the mantissa 0.9134 gives 8.19 from the antilog table, so the total is about 8,190 grams. Since $2^{13} = 8,192$, the answer agrees to four figures, with the small difference due to rounding in the four-figure table. (c) Long multiplication requires many digit-by-digit multiplications and carrying, which is slow and error-prone. The table has already done the hard work of converting numbers into exponents, so the farmer only has to look up two values, do one addition of small decimals, and look up one antilog. Converting multiplication into addition means a long multiplication takes about two minutes without a calculator.

Part 5: Powers, roots and answering the driving question

(a) A member of the Kericho Tea Growers SACCO borrows Ksh 50,000 at 4.5% compound interest per annum. The amount after 12 years is $50,000 \times 1.045^{12}$. Use the power law of logarithms and the table value $\log 1.045 = 0.0191$ to find the amount owed, to the nearest hundred shillings. Explain why the power law is true. (b) A surveyor in Nakuru County has a square plot of area 8,464 m². Use the root law and $\log 8,464 = 3.9276$ to find its side length. (c) Now write a final answer of about 6 to 8 sentences to the driving question: How can a Kericho tea farmer, or any Kenyan scientist, engineer or business person, use indices

(a) Power law: $\log(a^n) = n \log a$. This is true because a^n means a multiplied by itself n times, and by the product law the logarithm of a product is the sum of the logarithms, so adding $\log a$ n times gives $n \log a$. So $\log(1.045^{12}) = 12 \times 0.0191 = 0.2292$. The antilog of 0.2292 is about 1.696, so $1.045^{12} \approx 1.696$. Amount owed $\approx 50,000 \times 1.696 \approx$ Ksh 84,800. (b) Side = $\sqrt{8,464} = 8,464^{(1/2)}$. By the root law, $\log(\sqrt{a}) = (1/2) \log a$, so the log of the side is $3.9276 \div 2 = 1.9638$. The antilog of 1.9638 is 92.0, so the side length is about 92.0 m. (c) Index notation lets a Kericho tea farmer write repeated multiplication compactly: 2^3 g per plant for 2^{10} plants is $2^3 \times 2^{10} = 2^{13}$ g, because by the product rule the exponents add. Changing the exponent by 1 doubles or halves the harvest. A logarithm is just an exponent of 10, so $\log(ab) = \log a + \log b$ follows directly from the index law, which is why the farmer's colleague could add two small decimals from a table and get the same answer as multiplying two big numbers. The quotient, power and root laws extend the trick to division, compound interest (for example 1.045^{12} on a Ksh 50,000 loan) and square roots (for example the 8,464 m² plot in Nakuru). The four-figure table and antilog table turn each difficult operation into an addition, subtraction, or a multiplication or division of a small decimal, and in the end we recover the answer by looking up the antilog. In this way any Kenyan scientist, engineer or business person can calculate with very large or very small numbers quickly, without a calculator.

and logarithms to make sense of and calculate with numbers that are too large or too small for ordinary arithmetic? Use at least one index law, one logarithm law and the tea farm data in your answer.	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of index laws and notation	Correctly states and applies the product, quotient, zero, power, negative and fractional index laws with the tea-farm data, and explains why each law works by expanding repeated multiplication.	Applies most index laws correctly to the tea-farm numbers, with brief or partial explanation of why they work.	States some laws but makes errors in applying them, or gives answers without reasons or without reference to the tea farm.
Understanding of logarithms and antilogarithms	Defines the common logarithm as an exponent of 10, correctly identifies characteristic and mantissa, and accurately describes the four-step antilog procedure and why the antilog is the inverse of the log.	Gives the definition and handles characteristic and mantissa correctly, with minor gaps in explaining the antilog procedure.	Confuses logarithms with ordinary numbers, or cannot separate characteristic from mantissa, or cannot describe how the antilog works.
Explaining why log laws work and using them in calculations	Derives the product, quotient and power laws from the index laws, completes the calculations accurately with every step and law named, and the answers match the model values to the accuracy of the four-figure tables.	Completes the calculations with mostly correct steps and answers, with the derivation of the log laws partly explained.	Calculation steps are missing or incorrect, or the log laws are stated without any connection to the index laws.
Answering the driving question with Kenyan context	Gives a coherent, connected final answer that links index notation, the log laws and the four-figure tables, and uses the Kericho tea-farm data and at least one other Kenyan context accurately.	Gives a clear answer that uses the tea-farm data and at least one index law and one log law, with some connections missing.	Gives a general or incomplete answer that does not use the tea-farm data or fails to link indices to logarithms.

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